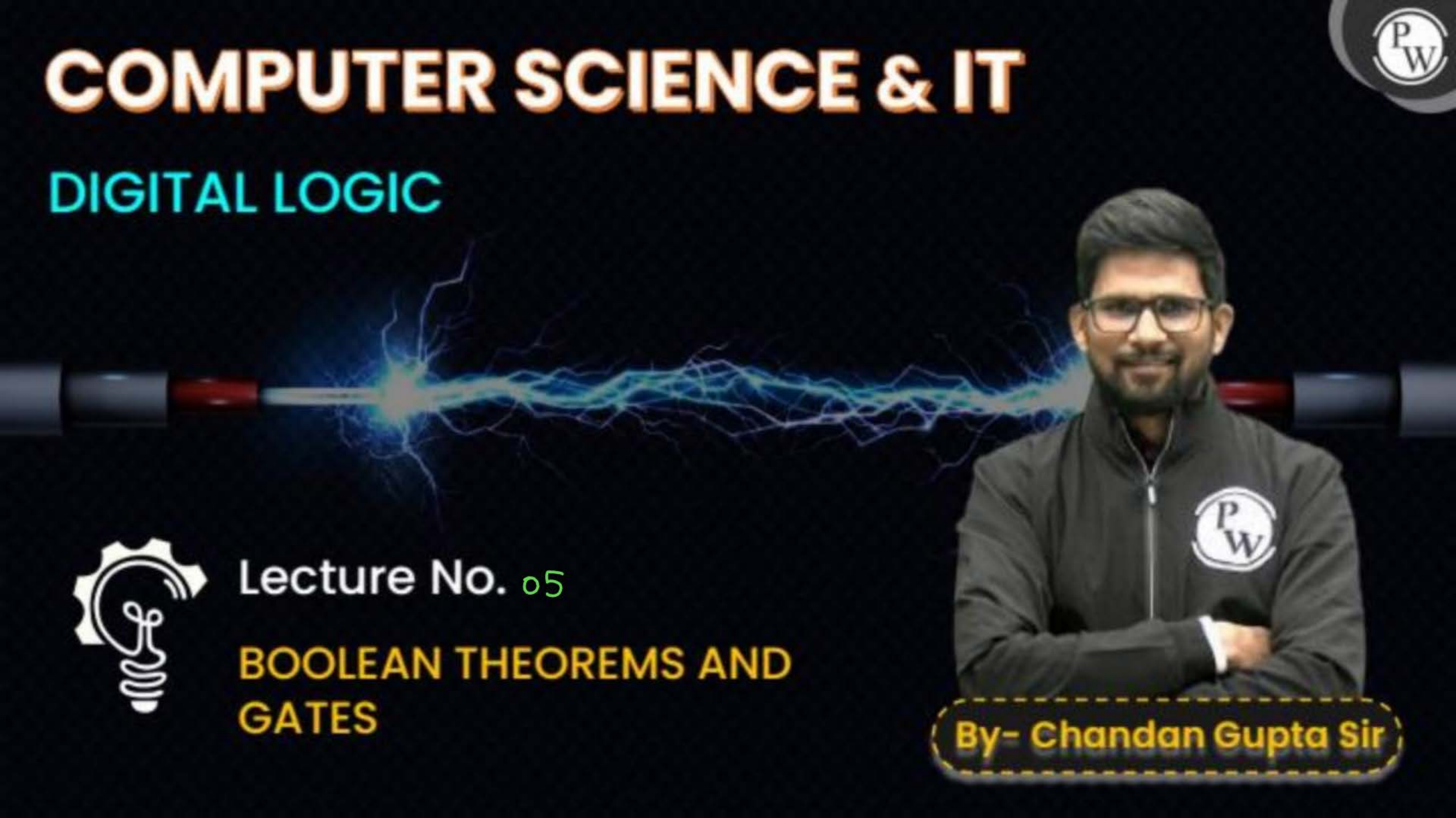
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 05

BOOLEAN THEOREMS AND GATES

By- Chandan Gupta Sir



NOT, AND, OR



XOR & XNOR gate

[XOR GATE]

'Exclusive OR' ⟶ gt is defined for two i/Ps only.

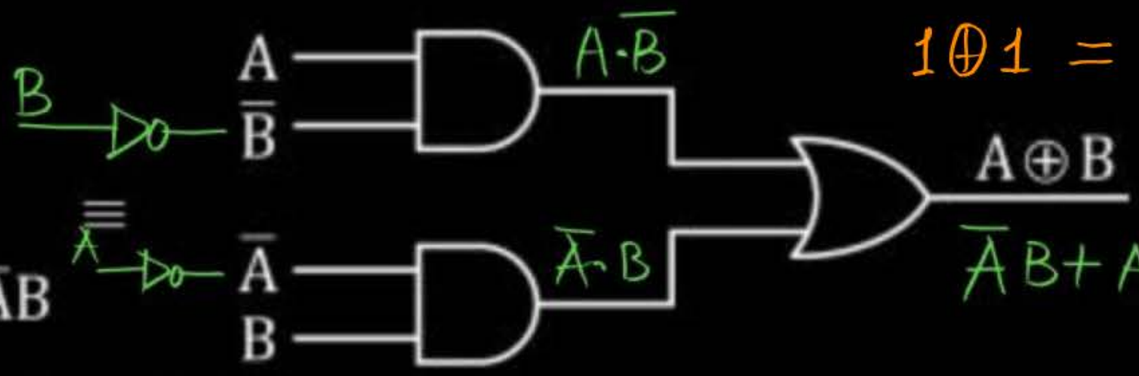
Representation :



$y = A \oplus B = A\bar{B} + \bar{A}B$

$0 \oplus 0 = \bar{0}\cdot 0 + 0\cdot\bar{0} = 0$

$0 \oplus 1 = \bar{0}\cdot 1 + 0\cdot\bar{1} = 1$



$= \bar{A}\cdot B + A\cdot\bar{B}$

$1 \oplus 0 = \bar{1}\cdot 0 + 1\cdot\bar{0} = 1$

$1 \oplus 1 = \bar{1}\cdot 1 + 1\cdot\bar{1} = 0$

$\bar{A}B + A\bar{B} = A \oplus B$

2^1 2^0

	A	B	$y = A \oplus B$
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	0

$y(A,B) = \Sigma(1,2) = \Pi(0,3)$

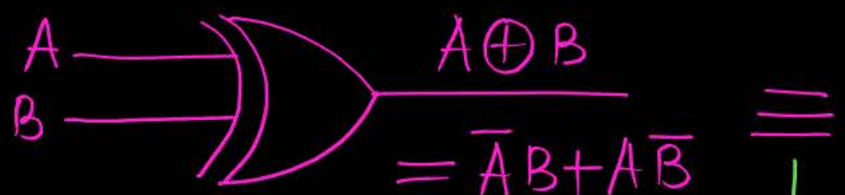
$= \bar{A}B + A\bar{B}$

$= (A+B)(\bar{A}+\bar{B})$

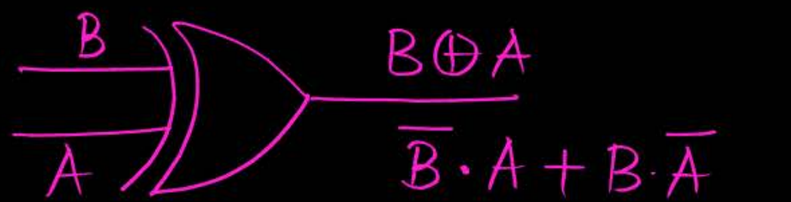
$= (\bar{A}B + A\bar{B})$

- $A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$
- $AB \oplus BC = \overline{AB} \cdot BC + AB \cdot \overline{BC}$
- $A \oplus (A+B) = \overline{A} \cdot (A+B) + A \cdot \overline{(A+B)}$
- $A\overline{B} \oplus CD = \overline{A\overline{B}} \cdot CD + (A\overline{B}) \cdot \overline{CD}$

- Commutative Law :



≡



$A \oplus B = B \oplus A$

It holds commutative law.

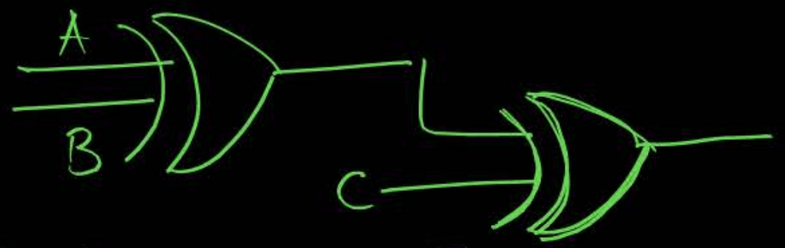
→ position of variables is irrelevant.

- **Associative Law :** It holds associative law → defined to calculate the multi i/p XOR operation.

$A \oplus B \oplus C \Longrightarrow (A \oplus B) \Longrightarrow (A \oplus B) \oplus C$ or $(A \oplus C) \oplus B$

$1 \oplus 0 \oplus 1 \Longrightarrow 1 \oplus 1 = 0$

$0 \oplus 0 = 0$



$A \oplus B \oplus C \oplus D \longrightarrow (A \oplus B) \Rightarrow (A \oplus B) \oplus C \Rightarrow [(A \oplus B) \oplus C] \oplus D$

$(2+3+5) =$

- Properties of XOR GATE :

- $A \oplus A = 0 = \bar{A} \cdot A + A \cdot \bar{A}$

  $\bar{A} \oplus \bar{A} = 0$

  $AB \oplus AB = 0$

- $A \oplus \bar{A} = 1 = \bar{A} \cdot \bar{A} + A \cdot \bar{\bar{A}}$

  $\bar{A} \oplus A = 1$

  $AB \oplus \overline{AB} = 1$

$A \oplus A = 0$

$A \oplus 0 = A = \bar{A} \cdot 0 + A \cdot \bar{0}$

$0 \oplus A = A$

$\bar{A} \oplus 0 = \bar{A}$

$AB \oplus 0 = AB$

$A \oplus \bar{A} = 1$

$1 \oplus A = \bar{A}$

$A \oplus 1 = \bar{A} = \bar{A} \cdot 1 + A \cdot \bar{1}$

$\bar{A} \oplus 1 = \bar{A} = A$

$AB \oplus 1 = \overline{AB}$

- Exchange properties of XOR GATE :

$A \oplus B = C \longrightarrow$ Given

$\Rightarrow A \oplus B = C$

$A \oplus C = B \longrightarrow$ true

$\Rightarrow A \oplus C = A \oplus A \oplus B = 0 \oplus B = B$

$\Rightarrow$

Similarly $B \oplus C = A$

$B \oplus C = B \oplus A \oplus B$

$B \oplus C = 0 \oplus A = A$



$A \oplus 0 = A$

$A \oplus 1 = \bar{A}$

Control i/p $= 0$

$= 1$ } both are enable i/ps

- $A \oplus A \oplus A \oplus A$ .....*n* times [*n* - represents no. of A]

  then o/p = 0 for *n*-even

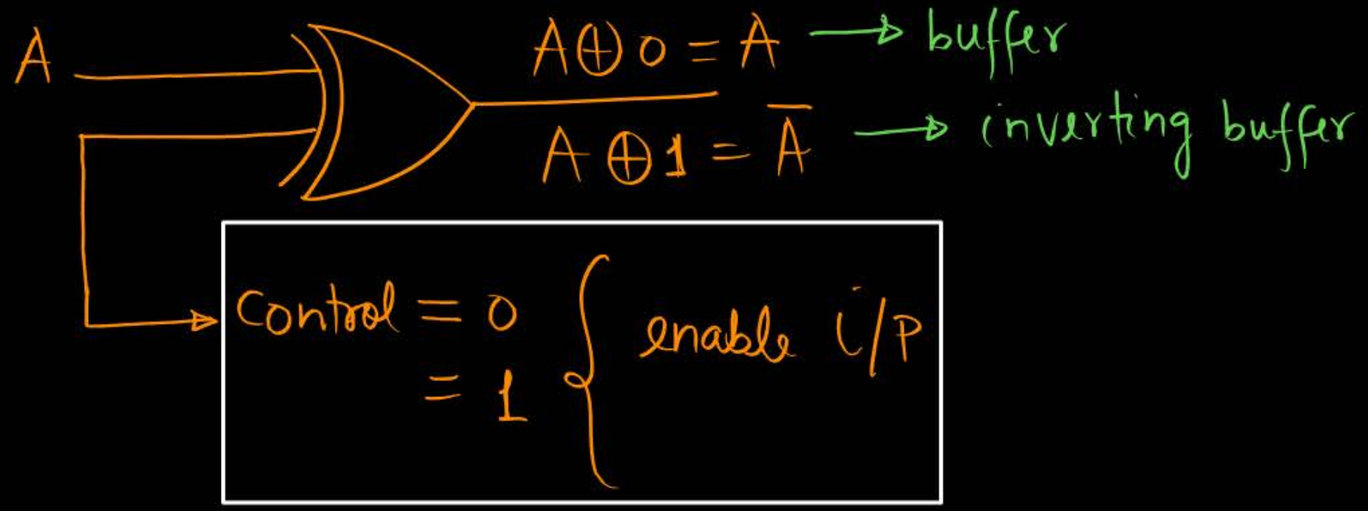
  = A for *n*-odd

$A \oplus A = 0$

$A \oplus A \oplus A \oplus A =$

$= 0 \oplus 0 = 0$

- **Buffer and inverting buffer using XOR :**



$A \oplus A \oplus A = 0 \oplus A$

$= A$

XNOR GATE

Exclusive NOR → (OR + NOT)

$0 \odot 0 = 1 \odot 1 = 1$

$0 \odot 1 = 1 \odot 0 = 0$

Representation :

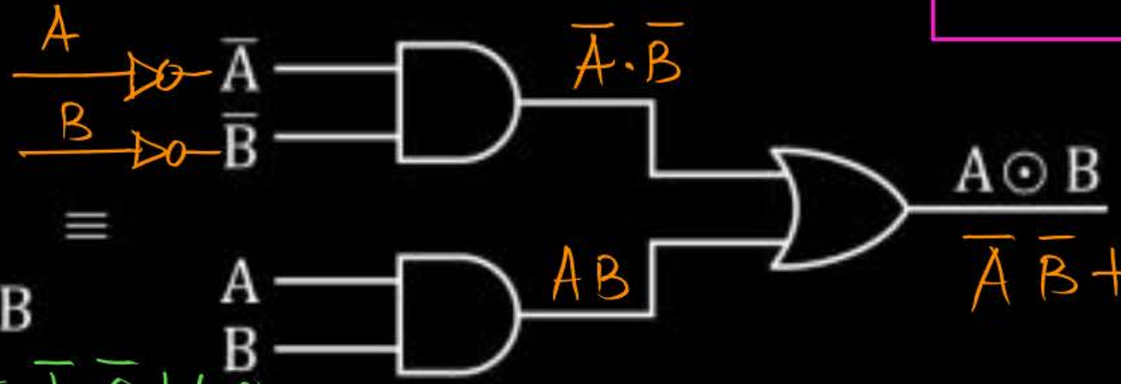


$A \odot B = \bar{A}\bar{B} + AB$

$0 \odot 0 = \bar{0}\cdot\bar{0} + 0\cdot 0 = 1$

$0 \odot 1 = \bar{0}\cdot\bar{1} + 0\cdot 1 = 0$

$2^1 \quad 2^0$



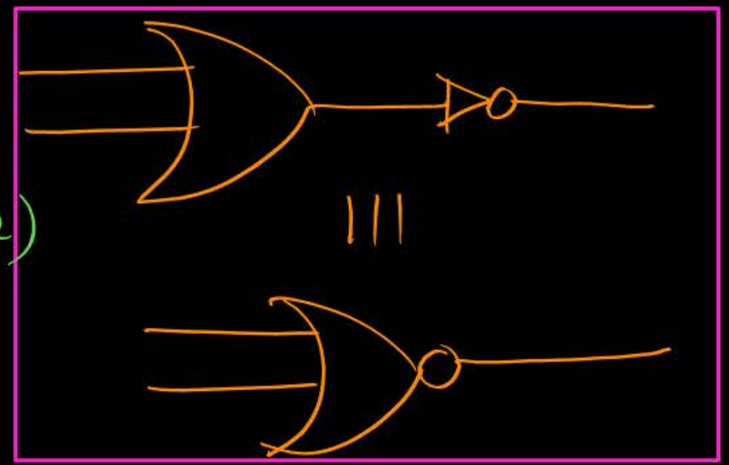
≡

$1 \odot 0 = \bar{1}\cdot\bar{0} + 1\cdot 0 = 0$

$1 \odot 1 = \bar{1}\cdot\bar{1} + 1\cdot 1 = 1$

	A	B	$y = A \odot B$	$Y_1 (A \oplus B)$
0	0	0	1	0
1	0	1	0	1
2	1	0	0	1
3	1	1	1	0

$y(A,B) = \Sigma(0,3) = \Pi(1,2)$



$\overline{A \oplus B} = (A \odot B)$

$A \oplus B = \overline{A \odot B}$

$= \bar{A}\cdot\bar{B} + AB = (A + \bar{B})(\bar{A} + B)$

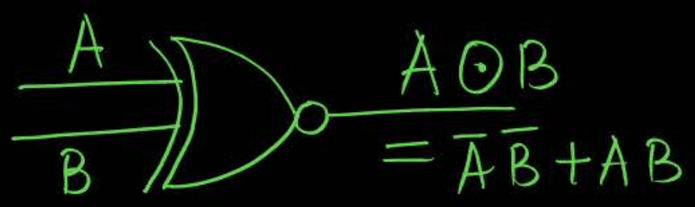
$$A \odot B = \overline{A} \cdot \overline{B} + A \cdot B$$

$$AB \odot BC = \overline{AB} \cdot \overline{BC} + AB \cdot BC$$

$$A \odot (A+B) = \overline{A} \cdot \overline{(A+B)} + A \cdot (A+B)$$

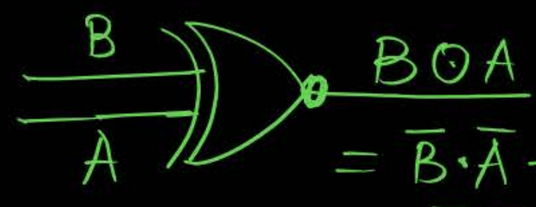
$$AB \odot CD = \overline{AB} \cdot \overline{CD} + AB \cdot CD$$

- Commutative Law :



$A \odot B = \bar{A}\bar{B} + AB$

$\equiv$



$B \odot A = \bar{B}\cdot\bar{A} + B\cdot A = \bar{A}\cdot\bar{B} + A\cdot B$

$A \odot B = B \odot A$

It holds commutative law

→ position of variables is irrelevant.

- **Associative Law :** It holds associative law ⟶ defined to calculate multi i/p XNOR.

$A \odot B \odot C \Rightarrow (A \odot B) \Rightarrow (A \odot B) \odot C$ or $(A \odot C) \odot B$

$1 \odot 0 \odot 1 \Rightarrow 0 \odot 1 = 0$

$1 \odot 0 = 0$

$A \odot B \odot C \odot D \Rightarrow (A \odot B) \odot (C \odot D)$ or $[(A \odot B) \odot C] \odot D$.

- Properties of XNOR gate :

- $A \odot A = 1 = \bar{A} \cdot \bar{A} + A \cdot A$

  $\bar{A} \odot \bar{A} = 1$

  $AB \odot AB = 1$

- $A \odot \bar{A} = 0 = \bar{A} \cdot \bar{\bar{A}} + A \cdot \bar{A}$

  $\bar{A} \odot A = 0$

  $AB \odot \overline{AB} = 0$

$A \odot A = 1$

$A \odot 1 = A = \bar{A} \cdot \bar{1} + A \cdot 1$

$\bar{A} \odot 1 = \bar{A}$

$AB \odot 1 = AB$

$A \odot \bar{A} = 0$

$A \odot 0 = \bar{A}$

$\bar{A} \odot 0 = A$

$AB \odot 0 = \overline{AB}$

- Exchange properties of XNOR gate :

$A \odot B = C$ ⟶ Given

$\Longrightarrow$ then $A \odot C = B$

then $C \odot B = A$ or $B \odot C = A$

Proof.

$A \odot C = A \odot A \odot B = 1 \odot B = B$

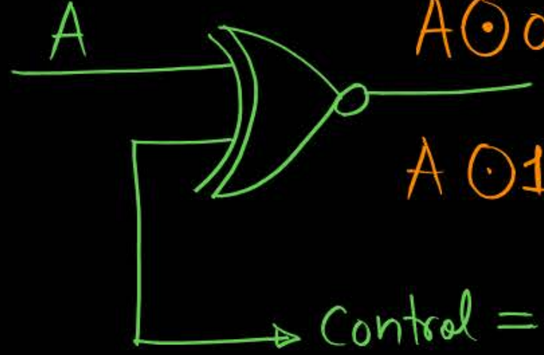
$B \odot C = B \odot A \odot B = 1 \odot A = A$

- $A \odot A \odot A \odot A$ .....*n* times [*n* - represents no. of A]

  then o/p = 1 for *n*-even

  = A for *n*-odd

- Buffer and inverting buffer using XNOR :



$A \odot 0 = \overline{A}$ → inverting buffer

$A \odot 1 = A$ → buffer

Control = 0 = 1 { both are enable i/Ps }

$A \odot A = 1$

$A \odot A \odot A \odot A$

$1 \odot 1 = 1$

$A \odot A \odot A = 1 \odot A$

$= A$

Combined IMP Properties of XOR and XNOR

- $\overline{A} \oplus B = \overline{\overline{A}} \cdot B + \overline{A} \cdot \overline{B} = A \odot B$

  $A \oplus \overline{B} = \overline{A} \cdot \overline{B} + A \cdot \overline{\overline{B}} = A \odot B$

  $\overline{A} \oplus \overline{B} = \overline{\overline{A}} \cdot \overline{B} + \overline{A} \cdot \overline{\overline{B}} = A \oplus B$

- $\overline{A} \odot B = \overline{\overline{A}} \cdot \overline{B} + \overline{A} \cdot B = A \oplus B$

  $A \odot \overline{B} = \overline{A} \cdot \overline{\overline{B}} + A \cdot \overline{B} = A \oplus B$

  $\overline{A} \odot \overline{B} = \overline{\overline{A}} \cdot \overline{\overline{B}} + \overline{A} \cdot B = A \odot B$

- $(\overline{A} \oplus \overline{B}) \oplus (C \oplus \overline{B}) \oplus (\overline{A} \oplus C) \oplus (\overline{A} \oplus \overline{B}) = (A \oplus B)$

$=$

$A \oplus A \oplus \overline{A} = \overline{A}$

$\overline{B} \oplus \overline{B} \oplus \overline{B} = 0 \oplus \overline{B} = \overline{B}$

$C \oplus C = 0$

$\overline{A} \oplus \overline{B} \oplus 0 = \overline{A} \oplus \overline{B} = A \oplus B$

- H.W.

Q.1. $A \oplus \bar{A} \oplus A \oplus \bar{A} \oplus A \oplus \bar{A} \oplus A = ?$

Q.2. $(A \oplus B \oplus \bar{C}) \oplus (\bar{A} \oplus B) \oplus (B \oplus \bar{C}) = ?$

Q.3. $1 \oplus A \oplus 0 \oplus \bar{A} \oplus \bar{B} \oplus 1 \oplus 0 = ?$

Q.4. $(\bar{A} \oplus \bar{B}) \oplus (\bar{B} \oplus \bar{C}) \oplus (\bar{A} \oplus \bar{C}) \oplus (A \oplus B \oplus C) = ?$

Q.5. $AB \oplus BC = ?$

Q.6. $AB \odot BC = ?$



2 Minute Summary

XOR & XNOR.

Thank you

GW

Soldiers !

